# NeST-Com: Fixed-Budget Next-Scale Token Communication

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***Abstract*—NeST-Com transmits images as next-scale residual tokens under a fixed per-image symbol budget. Actual-length frame fitting independently terminates arithmetic codes at each admissible within-scale prefix and selects the longest prefix that fits the protected frame body. The visual autoregressive (VAR) model uses known-token overwrite and recursive state updates to correct residuals and condition finer-scale predictions on received boundary tokens. On 500 ImageNet images at 1024 symbols and 10 dB, NeST-Com raises source-bit capacity utilization from 77.70% to 99.79% and reduces DISTS and DreamSim by 8.7% and 20.5% relative to a shared-model whole-scale baseline. Matched controls quantify the quality–computation tradeoff.**

***Index Terms*—Fixed symbol budget, image communication, next-scale tokens, perceptual reconstruction.**

## I. Introduction

Next-scale representations describe an image as complete coarse maps followed by residual refinements at finer scales [1]. Each scale spans the full latent field and conditions the next finer scale, yielding a progressively refinable description and a generative prior for untransmitted content.

Under a fixed per-image symbol budget, transmitting only complete scale maps leaves capacity unused whenever the next residual map exceeds the remaining capacity, even if a prefix of its tokens would fit.

ARPC selects scales for progressive image compression [2]. Ada-TokenCom jointly adapts prefix length and the modulation and coding scheme (MCS) under a long-term symbol budget [3], and ResARC codes residuals to compensate for suffix-generation errors [4]. Deep joint source–channel coding (JSCC) methods such as SwinJSCC map images directly to channel symbols and serve as system-level baselines [5]. For a fixed digital frame, the open question is how to terminate a decodable prefix within a scale and use the recovered boundary tokens at the receiver.

This letter proposes NeST-Com. Its frame-fitting rule uses a coder-specific nondecreasing-length property to select the longest independently terminated within-scale prefix whose arithmetic-code length fits the protected frame body. The receiver incorporates the recovered boundary tokens through VAR's known-token overwrite to correct residuals and condition finer-scale predictions. Experiments on 500 ImageNet images and matched controls examine how capacity utilization and boundary-token use contribute to the gains and quantify the quality–computation tradeoff.

## II. NeST-Com Framework

### A. System Model and Next-Scale Representation

We transmit $x \in [0,1]^{256\times256\times3}$ using $N$ complex symbols over AWGN,

$$y = s + n, \quad n \sim \mathcal{CN}(0, \sigma^2 I_N), \quad \mathrm{SNR} = 10\log_{10}(E_s/\sigma^2), \tag{1}$$

with nominal $E_s = 2$. TX uses the frozen policy for its design SNR. RX uses the actual noise variance for LLRs. Digital constellations have average energy two without per-frame renormalization. A terminated rate-$1/2$ convolutional code protects the 12-bit profile ID and CRC16 in a fixed 68-symbol header. The remaining $N - 68$ symbols carry the LDPC-coded body with public frame-dependent scrambling.

The frozen VAR tokenizer produces ten residual token maps $q_s$, with side lengths $(1, 2, 3, 4, 5, 6, 8, 10, 13, 16)$ and a 4096-entry codebook [1]. Let $L_s$ denote the number of positions at scale $s$, and let $C_s$ denote codebook lookup followed by upsampling to the full latent grid and the scale-dependent residual transform. The latent sum $\sum_s C_s(q_s)$ progressively refines a full-field description. The decoder $D_c$ is fine-tuned once and then frozen. Because the token probabilities at each scale depend only on complete coarser maps, RX can reproduce the transmitter's integer cumulative distribution functions (CDFs, total frequency $2^{24}$) and decode any independently terminated prefix.

### B. Within-Scale Endpoints and the Nondecreasing-Length Rule

An endpoint $e = (m, K)$ retains complete coarse maps and a raster-ordered boundary prefix:

$$\pi_e = (q_1, \ldots, q_m, q_{m+1,1:K}). \tag{2}$$

For $K = 0$, including $e = (10, 0)$, the boundary prefix is empty. Raster order specifies positions without a mask. WHOLE permits $(m, 0)$ for $m = 4, \ldots, 10$. NeST permits every integer $K = 0, \ldots, L_{m+1} - 1$ for $m = 4, \ldots, 9$, together with $(10, 0)$: 651 endpoints from 30 through 680 retained tokens. Quarter/Eighth retain $K = \lfloor jL_{m+1}/d \rfloor$, $j = 1, \ldots, d-1$, for $d = 4/8$, plus complete-scale boundaries, with duplicates removed. Token is the main configuration.

For MCS $a = (M, r)$, the coded body has $n_a = (N - 68)\log_2 M$ bits and information length $k_a = \lfloor rn_a \rfloor$. Let $b(x, e)$ denote the actual bit length obtained by independently terminating the arithmetic code for token prefix $\pi_e$. Feasibility requires

$$b(x, e) + 29 \le k_a. \tag{3}$$

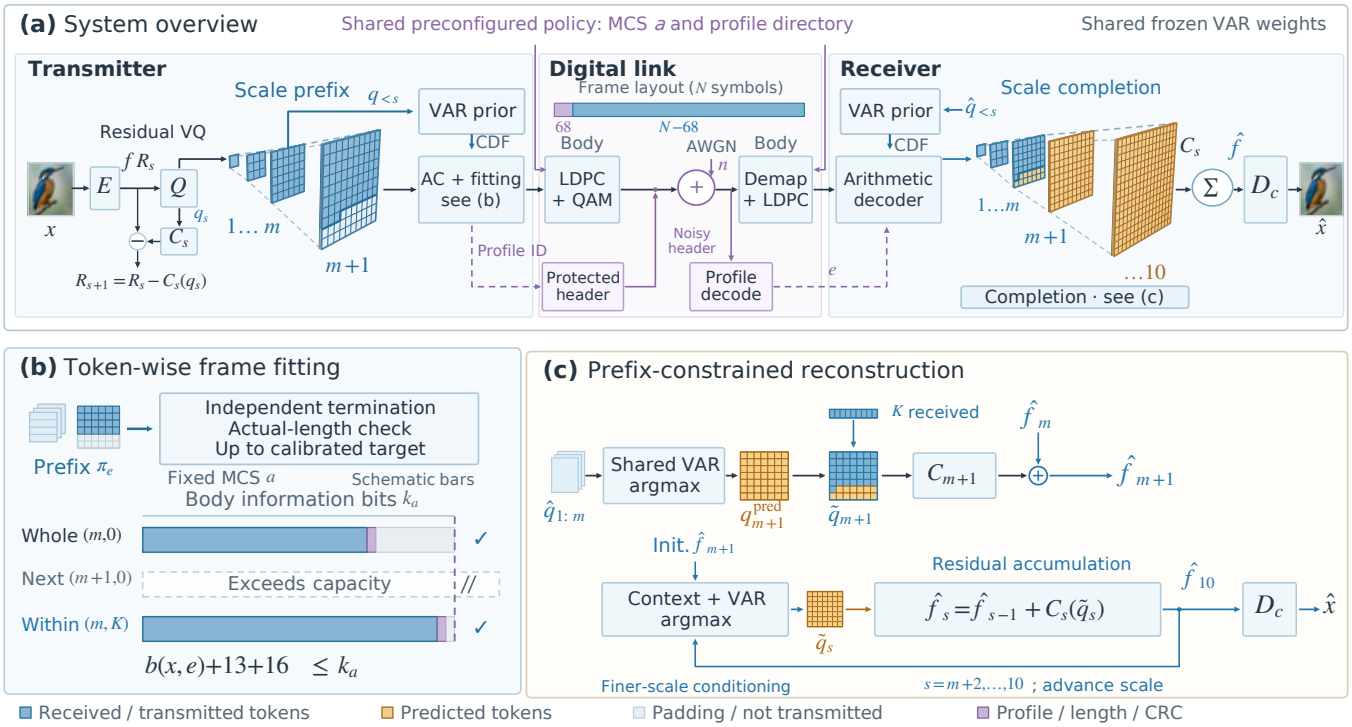



Fig. 1. NeST-Com. (a) The shared policy fixes the MCS and profile directory; the protected Profile ID selects the endpoint. VAR and known coarse tokens provide coding probabilities. (b) Independently terminated prefix bitstreams up to the target endpoint fit the body capacity; bars are schematic. (c) Received tokens overwrite boundary predictions. Recursive state updates accumulate residuals and condition each finer scale before decoding. Unreceived boundary positions retain their coarse-conditioned predictions.

The source-bit capacity utilization is $\eta(x,e) = b(x,e)/(k_a - 29)$, where $k_a - 29$ is the capacity available to the source payload. The 29 bits comprise a 13-bit length and CRC16. Remaining positions are zero-padded before 5G NR LDPC coding, rate matching [6], and QAM.

A public method/MCS directory, frozen before calibration and shared across images, maps source profiles to endpoints and includes a NULL profile. One traversal shares scale-wise probability evaluation and incremental coding. Non-mutating termination produces independent prefix bitstreams, not slices of a longer codeword.

The fitting rule rests on a coder-specific property. For the fixed token/CDF traversal and non-mutating finish used here, the actual length after $t$ tokens is $b_t = E_t + P_t + 2$, where $E_t$ and $P_t$ count emitted and pending bits. Renormalization increments $E_t + P_t$. Other interval updates preserve it. Thus $b_t$ is nondecreasing, and for a fixed MCS and ceiling the feasible endpoints form an initial segment. TX selects the longest feasible prefix in a single scan of the candidate lengths. If no source prefix fits, it sends a NULL frame with a two-bit zero payload over the full $N$-symbol budget.

### C. Prefix-Constrained Reconstruction

RX returns a fixed gray image if a profile or body CRC fails, the profile is absent from the directory, or the declared length, padding or arithmetic stream is invalid. Valid NULL frames also produce gray, and both outcomes remain in all metric denominators. Otherwise, RX decodes the received prefix using the shared CDFs. At $e = (10, 0)$, all maps are received and directly decoded. For an incomplete prefix, RX predicts the boundary map by argmax conditioned on the complete received maps and overwrites the received positions according to

$$\tilde{q}_{m+1,i} = \begin{cases} \hat{q}_{m+1,i}, & i \le K, \\ q^{\text{pred}}_{m+1,i}, & i > K. \end{cases} \tag{4}$$

RX then generates the finer maps from the merged boundary map and decodes the accumulated residuals as

$$\hat{x} = D_c\left(\sum_{s=1}^{10} C_s(\tilde{q}_s)\right). \tag{5}$$

Received boundary tokens correct the residual representation and, through the merged boundary map, condition finer-scale prediction. Unreceived boundary positions retain their original coarse-conditioned predictions without a same-scale prediction loop. At the last scale, overwrite changes only the residual contribution. Supplement S1 gives the full algorithm, coder-specific proof, decoder training and implementation details.

## III. Experimental Results

### A. Setup and Calibration Protocol

We test 500 ImageNet validation images [7], disjoint from 1000 calibration images, at $256 \times 256$: $N = 1024$ at 4/7/10/19 dB and $N = 512$ at 4/10/19 dB. We average three realizations per image, including gray failures, then compute equal-source means and direct paired 95% intervals using 10 000 source resamples. Controls use a fixed 100-image subset. Calibrated

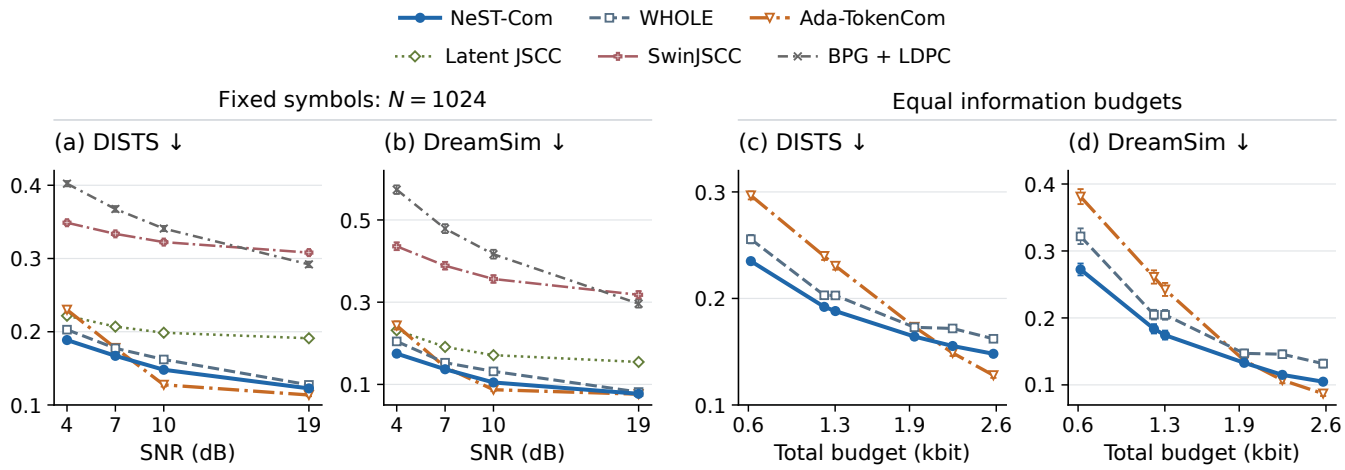



Fig. 2. Source-referenced reconstruction on 500 ImageNet images. (a,b) Six methods at $N = 1024$; WHOLE, NeST and Ada share the 12-MCS calibration procedure. Three outcomes per source include gray failures. (c,d) Equal, error-free information budgets pay 57 shared protocol bits and Ada's additional 10-bit class; WHOLE/NeST share ceiling $(10, 0)$. Points and bars are equal-source means and pointwise 95% source-bootstrap intervals after within-source averaging. Ada-TokenCom is the predicted-class, fixed-frame adaptation. Lower values are favorable.

token systems and matched controls share noise; external systems retain their original noise.

Latent JSCC maps tokenizer latents to symbols and shares $D_c$ (MSE, LPIPS and latent-consistency losses, 100k updates). SwinJSCC [5] uses RGB MSE (200k updates, calibration-MSE selection). Both train on 20 000 images at $N = 1024$. BPG [8] uses LDPC/QAM. Ada-TokenCom [3] is a predicted-class, fixed-frame adaptation: LlamaGen-L/VQ16, prefix arithmetic coding, CFG=4, $T = 1$, top-$k = 2000$, top-$p = 1$, and a paid 10-bit ConvNeXt class. Long-term control and ARQ are excluded.

DISTS [9], DreamSim [10], PSNR, LPIPS-Alex [11] and CLIP-I [12] use native float RGB with fixed preprocessing. Timing uses 32 images, two warmups, three measurements, batch one, one RTX PRO 6000 and two physical CPU cores. We separately measure GPU-synchronized TX (RGB-to-symbols) and RX (symbols-to-RGB) processing times. They include coding, link processing and generation, excluding loading, I/O and scoring; total processing time is their sum. Supplementary Sections S2–S8 give complete results, controls, examples, timing and transfer details, including ARPC's specified digital-link adaptation.

Policies are calibrated once and then frozen. At $N = 1024$, WHOLE, NeST and Ada share 12 MCSs combining QPSK, 16-QAM or 64-QAM with code rates $1/2$, $2/3$, $3/4$ or $5/6$. WHOLE has seven ceilings, NeST uses 25 Quarter ceilings with Token fallback, and Ada allows prefixes through 256 tokens. All 396 candidates are screened on 100 calibration images with one noise realization. For each method and SNR, the top three candidates and a preset QPSK rate-$1/2$ anchor advance to 1000 calibration images with three realizations each. The highest equal-source mean DINOv2 cosine similarity [13], including gray failures, determines the policy. The $N = 512$ policies are calibrated separately (Table IV).

### B. Within-Scale Fitting and Mechanism Attribution

Table I(C) isolates the receiver's use of boundary tokens by holding the received NeST prefixes fixed. A0 predicts the boundary and finer maps from the complete coarse maps. B0 applies the received tokens to correct the boundary residual while retaining A0's finer-scale conditioning. C0 also uses the corrected boundary map to condition the finer scales. At 10 dB, B0 improves PSNR by 0.287 dB over A0, and C0 adds 0.537 dB over B0. The paired PSNR, DISTS and DreamSim intervals favor both steps.

In the 500-image comparison, NeST and WHOLE share the model, coder and receiver and use separately calibrated policies. At $N = 1024$ and 10 dB, source-bit capacity

TABLE I
WITHIN-SCALE TRANSMISSION. (A) SHARED-MODEL NEST–WHOLE DIFFERENCES. (B) MATCHED ENDPOINT GRANULARITY. (C) ORDERED RECEIVER INTERVENTIONS. BRACKETS ARE SOURCE-PAIRED 95% INTERVALS; SAMPLE SIZES ARE SPECIFIED FOR EACH BLOCK.

*A. 500 sources, $N = 1024$, uniformly calibrated policies*

| SNR | Util. W→N (%) | ΔDISTS | ΔDreamSim |
|---|---|---|---|
| 4 | 73.81→99.56 | -0.0142 [-0.0155, -0.0127] | -0.0297 [-0.0330, -0.0264] |
| 7 | 83.19→99.71 | -0.0103 [-0.0139, -0.0068] | -0.0159 [-0.0207, -0.0112] |
| 10 | 77.70→99.79 | -0.0141 [-0.0153, -0.0128] | -0.0270 [-0.0299, -0.0241] |
| 19 | 85.24→99.83 | -0.0048 [-0.0053, -0.0043] | -0.0043 [-0.0051, -0.0036] |

*B. 100 quality sources; 32 timing sources; ceiling $(8, 126)$*

| Grid | Util. (%) | DISTS | DreamSim | TX (ms) |
|---|---|---|---|---|
| Whole | 78.13 | 0.1650 | 0.1387 | 185.2 |
| Quarter | 95.19 | 0.1535 | 0.1143 | 215.8 |
| Eighth | 97.44 | 0.1516 | 0.1137 | 217.8 |
| Token | 99.85 | 0.1505 | 0.1115 | 248.4 |

*C. 100 sources, 10 dB, same received prefixes*

| Intervention | ΔPSNR (dB) | ΔDISTS | ΔDreamSim |
|---|---|---|---|
| B0–A0 | +0.287 [0.240, 0.335] | -0.0062 [-0.0077, -0.0048] | -0.0113 [-0.0147, -0.0083] |
| C0–B0 | +0.537 [0.404, 0.682] | -0.0083 [-0.0102, -0.0065] | -0.0159 [-0.0205, -0.0116] |

A0: prior completion without received boundary tokens. B0: A0 plus boundary residual correction. C0: B0 plus updated finer-scale conditioning.




Fig. 3. Selected reconstructions at $N = 1024$ and 10 dB. WHOLE, NeST and the predicted-class fixed-frame Ada-TokenCom adaptation use the policies in Table IV. Values are full-image DISTS↓ / DreamSim↓ for each method's first prescribed realization. The sources were selected for favorable perceptual examples before recalibration.

TABLE II
AUXILIARY QUALITY ON 500 TEST IMAGES AT $N = 1024$, 10 DB. MEANS INCLUDE THREE REALIZATIONS PER SOURCE AND GRAY FAILURES.

| Method | PSNR (dB)↑ | LPIPS↓ | CLIP-I↑ |
|---|---|---|---|
| WHOLE | 19.31 | 0.199 | 0.858 |
| NeST-Com | 20.19 | 0.166 | 0.877 |
| Ada-TokenCom | 19.56 | 0.167 | 0.884 |
| Latent JSCC | 20.80 | 0.173 | 0.789 |
| SwinJSCC | 23.03 | 0.469 | 0.669 |
| BPG + LDPC | 22.54 | 0.589 | 0.635 |

utilization rises from 77.70% to 99.79%, mean DISTS falls from 0.1621 to 0.1481, and mean DreamSim falls from 0.1317 to 0.1048. Table I(A) reports the paired NeST–WHOLE differences computed before rounding, $-0.0141$ for DISTS and $-0.0270$ for DreamSim. At the same operating point, mean PSNR increases from 19.31 dB for WHOLE to 20.19 dB for NeST, a gain of 0.88 dB (Table II). All five paired metrics favor NeST at all seven budget/SNR points (Supplement S2).

Table I(B) isolates granularity on a separate 100-image quality subset by fixing the model, 16-QAM rate 2/3, ceiling $(8, 126)$, fitting rule and noise. Eighth retains approximately 92% of the Whole-to-Token gain in PSNR, DISTS and DreamSim. Token raises utilization from 97.44% to 99.85%, adding $0.066\,[0.034, 0.101]$ dB PSNR, with DISTS/DreamSim differences $-0.0011\,[-0.0020, -0.0002]$/$-0.0022\,[-0.0042, -0.0001]$, at 30.7 ms (14.1%) additional TX cost. Eighth offers lower cost; Token prioritizes fuller capacity use and the additional measured quality gain.

### C. System-Level Comparison

Fig. 2(a,b) compares six methods at equal symbol budgets. NeST has lower DISTS and DreamSim than SwinJSCC, Latent JSCC and BPG at all four $N = 1024$ SNRs (24 paired intervals excluding zero). SwinJSCC and BPG have higher PSNR.

TABLE III
REDUCED-BUDGET RECONSTRUCTION ON 500 SOURCES AT $N = 512$. MEANS INCLUDE ALL FAILURES; PSNR DIFFERENCES USE DIRECT SOURCE-PAIRED 95% INTERVALS.

| SNR / method | PSNR↑ | DISTS↓ | DreamSim↓ |
|---|---|---|---|
| 4 / WHOLE | 15.11 | 0.2580 | 0.3247 |
| 4 / NeST-Com | 15.61 | 0.2363 | 0.2740 |
| 4 / Ada-TokenCom | 11.32 | 0.2996 | 0.3847 |
| NeST–WHOLE ΔPSNR: +0.504 [0.440, 0.573] dB | | | |
| 10 / WHOLE | 16.82 | 0.2030 | 0.2049 |
| 10 / NeST-Com | 17.41 | 0.1922 | 0.1837 |
| 10 / Ada-TokenCom | 13.16 | 0.2367 | 0.2582 |
| NeST–WHOLE ΔPSNR: +0.592 [0.544, 0.643] dB | | | |
| 19 / WHOLE | 18.64 | 0.1718 | 0.1458 |
| 19 / NeST-Com | 19.63 | 0.1554 | 0.1150 |
| 19 / Ada-TokenCom | 17.69 | 0.1496 | 0.1078 |
| NeST–WHOLE ΔPSNR: +0.994 [0.940, 1.048] dB | | | |

Against Ada, NeST improves both distances at 4 dB and DISTS at 7 dB. Ada has lower distances at 10 dB and lower DISTS at 19 dB. NeST has higher PSNR at all four SNRs. At 7 dB, WHOLE, NeST and Ada reject 11, 7 and 10 of 1500 receptions, respectively. NeST rejects one frame at 4 dB. All other token-system points have zero rejections, and no valid NULL frames occur. Table II gives auxiliary metrics at 10 dB.

Fig. 2(c,d) compares the schemes at six equal, error-free information budgets (620–2577 bits), including 57 shared protocol bits and Ada's additional 10-bit class. NeST improves all five metrics over WHOLE. Against Ada, DISTS favors NeST at the four smallest budgets and DreamSim at the first three. Ada has lower values on both metrics at 2248 and 2577 bits.

### D. Reduced Budget, Transfer and Threshold Behavior

The $N = 512$, 10-dB granularity control fixes 100 images, three shared noise realizations, 16-QAM at rate $2/3$, ceiling $(7, 0)$, model and receiver. Eighth retains 87% of the Whole-to-

TABLE IV
FROZEN POLICIES: MCS ENTRIES ARE (BITS/SYMBOL, CODE RATE); $e$ IS THE TARGET CEILING, WITH FEASIBLE FALLBACK. ADA'S CEILING IS 256 TOKENS. BUDGETS ARE CALIBRATED INDEPENDENTLY.

| $N$/SNR | $a_{\mathrm{W}}$ | $e_{\mathrm{W}}$ | $a_{\mathrm{N}}$ | $e_{\mathrm{N}}$ | $a_{\mathrm{Ada}}$ |
|---|---|---|---|---|---|
| 1024/4 | $2,2/3$ | $(7,0)$ | $2,2/3$ | $(7,25)$ | $2,2/3$ |
| 1024/7 | $4,1/2$ | $(8,0)$ | $4,1/2$ | $(8,0)$ | $4,1/2$ |
| 1024/10 | $4,2/3$ | $(9,0)$ | $4,2/3$ | $(8,126)$ | $4,2/3$ |
| 1024/19 | $6,5/6$ | $(10,0)$ | $6,5/6$ | $(10,0)$ | $4,5/6$ |
| 512/4 | $2,2/3$ | $(6,0)$ | $2,2/3$ | $(5,27)$ | $2,2/3$ |
| 512/10 | $4,2/3$ | $(7,0)$ | $4,2/3$ | $(7,0)$ | $4,2/3$ |
| 512/19 | $6,5/6$ | $(8,0)$ | $6,5/6$ | $(8,84)$ | $6,5/6$ |

Token PSNR gain. Token further improves PSNR, LPIPS and DreamSim, with DISTS and CLIP-I intervals crossing zero.

On the 500 test images at $N = 512$, all five paired intervals favor NeST over WHOLE at all three SNRs. PSNR gains range from 0.50 to 0.99 dB (Table III). At 10 dB both methods use ceiling $(7, 0)$ and 16-QAM at rate $2/3$, and utilization rises from 79.30% to 99.54%.

We evaluate 24 Kodak center crops using ImageNet-calibrated policies, with 64-QAM at code rate $5/6$ for Ada at 19 dB. NeST improves PSNR, DISTS and DreamSim over WHOLE at all three SNRs, and all 864 receptions succeed.

Near the channel threshold, the three token systems use fixed 10-dB timing policies, and RX uses the actual noise variance. Testing 100 images with 20 noise realizations per image at each of five SNRs from 8 to 10 dB gives 30 000 receptions. All three methods reject every frame at 8 dB. At 9 dB, rejection rates are 15.75%, 14.95% and 13.50% for WHOLE, NeST and Ada, respectively. They fall to 0.10–0.20% at 9.5 dB and to zero at 10 dB. All rejected gray outputs remain in the metric denominators.

Fig. 3 illustrates NeST preserving truck markings and foreground contours. These selected cases favor NeST in DISTS and DreamSim and SwinJSCC in PSNR.

We measure processing time for each token system at $N = 1024$ and 10 dB using 16-QAM at code rate $2/3$ and the corresponding source-prefix ceiling in Table IV. TX and RX take 248.44 and 403.99 ms for NeST, respectively, and 215.29 and 366.42 ms for WHOLE. Total processing time is 652.43 ms for NeST and 39.17 ms for SwinJSCC. The 3.064 s measured for compiled Ada is paired with its own 32-image quality evaluation. The main Ada quality results use the original implementation.

## IV. CONCLUSION

NeST-Com pairs a frame-fitting rule with a receiver that uses the recovered boundary tokens. A coder-specific nondecreasing-length property allows the transmitter to select the longest independently terminated prefix that fits the frame. At the receiver, the recovered tokens overwrite boundary predictions, correcting residuals and updating the context for finer-scale prediction. On 500 ImageNet images at $N = 1024$ and 10 dB, NeST-Com raises source-bit capacity utilization from 77.70% to 99.79% and reduces DISTS and DreamSim by 8.7% and 20.5% relative to the shared-model whole-scale baseline. The Eighth grid offers lower transmitter cost with a small quality loss relative to Token. This tradeoff suits image transmission under tight symbol budgets when the application prioritizes perceptual quality and can accommodate the measured processing time.

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